

KG RAG ASSIGNMENT

Assignment 1: Hospital Q&A System with Knowledge Graph RAG

TASKS

Take the “hospital visits” pdf and “disease treatments” pdfs (use all the 4 PDFs) and:

1) Take it through the Graph **indexing** flow

Upload the PDFs to Neo4j Graph Builder to directly create graph from pdf

2) Implement app in Python that will take a user question and use LLM to generate cypher query based on the question

Hint: Pass the llm schema to the LLM in context

3) Send the cypher query to neo4j and obtain the results

4) send the results to LLM to generate final response.

Demonstrate with few questions such as:

1) What are the treatments for <disease>

2) Which doctors did patient X visit

3) Which diseases does drug Y treat

And more (at least 5)

Submit: Link to colab notebook demonstrating the above

Assignment 2: Compare graph DB outputs with Vector database

TASKS

Take the same datasets from above and implement a RAG solution with vector database

Use the same prompts above and compare responses.

Submit: Link to colab notebook demonstrating the above

Assignment 3: Hybrid RAG with Vector DB and Graph DB using Langchain

TASKS

For the same data and schema, implement a hybrid indexer and retriever using Langchain

Try the prompts from assignment 1 and see if there is any improvement

Assignment 4: Hybrid RAG with Vector DB and Graph DB using Llamaindex

TASKS

For the same data and schema, implement a hybrid indexer and retriever using Llamaindex

Try the prompts from assignment 1 and see if there is any improvement